Akoizo Original Practice - Form 2 - Answer key

Division B | Experimental Design | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Which is a controlled variable rather than a control treatment?

Unit: Problem, hypothesis, variables, and experimentation.
Worked-case data: All low-level trials occur early and all high-level trials occur late while the instrument drifts upward.

A. A baseline reference level

B. An untreated comparison

C. Keeping light constant

D. A no-added-fertilizer group

Correct: C

2. (2 points) What conclusion is supported by these trials?

Unit: A clear Division B investigation report.
Worked-case data: 25 cm^2 towel: Absorbed masses: 2, 3, 4 g. 50 cm^2 towel: Absorbed masses: 5, 6, 7 g. Controls: Same towel brand, soak time, and draining method.
Separate investigation observations:
25 cm^2 towel: Absorbed masses: 2, 3, 4 g.
50 cm^2 towel: Absorbed masses: 5, 6, 7 g.
Controls: Same towel brand, soak time, and draining method.

A. The larger towel absorbed less water

B. Soak time caused the change because it was increased

C. Every individual measurement doubled

D. Doubling area increased mean absorbed mass from 3 g to 6 g

Correct: D

3. (2 points) Standard deviation has…

Unit: Quantitative data, graphs, and statistics.
Worked-case data: For response values 8, 10, 12, calculate mean, range, and sample SD.

A. The independent-variable units necessarily

B. The response's units

C. Squared response units

D. No units always

Correct: B

4. (2 points) A failed hypothesis means the experiment is invalid?

Unit: The experiment and the prompt packet.
Worked-case data: A prompt supplies a ramp, object, ruler, and timer. Turn "study motion" into a testable question.

A. Only if data vary

B. Only if the graph is flat

C. No

D. Yes

Correct: C

5. (2 points) A conclusion should answer…

Unit: Errors, conclusions, and recommendations.
Worked-case data: Every timer reading is 0.5 s too high. Would ten more repeats eliminate the bias?

A. The original measured question

B. Every possible mechanism

C. A preferred hypothesis regardless of data

D. A different untested outcome

Correct: A

6. (4 points) Improve "release and time it."

Unit: Materials, reproducible procedures, and qualitative data.
Worked-case data: A trial has no timer reading because the observer missed the endpoint. What should the table and procedure record?

answer: Specify release location and method, start/stop marks, timer operation, reset, trial count, and recording units. (4 points).

7. (4 points) Name three dependent reporting tasks.

Unit: Practice, timing, teamwork, and final review.
Worked-case data: A team finishes data collection with little time left and no graph or analysis. Suggest a targeted practice change.

answer: Statistics, graphing, and conclusions depend on collected measurements; the abstract depends on completed analysis. (4 points).

8. (4 points) Give a variance mechanism for timing.

Unit: Claims, evidence, and reasoning for trends and outliers.
Worked-case data: At levels 1,2,3, means are 12,16,20 units, with SD 2 at each level. Write a bounded trend CER.

answer: Inconsistent start/stop reaction can vary measured duration across trials, increasing spread. (4 points).

9. (4 points) Why not invent qualitative notes after analysis?

Unit: Materials, reproducible procedures, and qualitative data.
Worked-case data: A trial has no timer reading because the observer missed the endpoint. What should the table and procedure record?

answer: They would not be genuine observations and could falsely justify selective exclusion or an unsupported mechanism. (4 points).

10. (4 points) What should a changed procedure record?

Unit: Practice, timing, teamwork, and final review.
Worked-case data: A team finishes data collection with little time left and no graph or analysis. Suggest a targeted practice change.

answer: Version, affected trials, reason, and whether combining data remains justified. (4 points).

11. (4 points) What limits extrapolation?

Unit: Claims, evidence, and reasoning for trends and outliers.
Worked-case data: At levels 1,2,3, means are 12,16,20 units, with SD 2 at each level. Write a bounded trend CER.

answer: Only the tested range and conditions were observed; the relationship may change outside them. (4 points).

12. (6 points) Solve this case and explain your reasoning: All low-level trials occur early and all high-level trials occur late while the instrument drifts upward.

reasoning: Identify time and level as systematically associated.
Upward drift can imitate or exaggerate a positive level effect.
Randomize order or use suitable blocking and calibration checks. (3 points).

conclusion: Replication alone does not remove systematic confounding. (3 points).

13. (2 points) Define “Trial” in the context of Experimental Design.

answer: One repetition of a measurement. (2 points).

14. (2 points) Define “Claim” in the context of Experimental Design.

answer: Answer to the defined question. (2 points).

15. (2 points) Define “Controlled variable” in the context of Experimental Design.

answer: Condition kept comparable. (2 points).

16. (2 points) Define “Coherence” in the context of Experimental Design.

answer: Agreement among design, data, analysis, and conclusions. (2 points).

17. (2 points) Define “Reasoning” in the context of Experimental Design.

answer: Explanation connecting evidence to a claim. (2 points).

18. (2 points) Define “Precision” in the context of Experimental Design.

answer: Consistency or limited spread of measurements. (2 points).